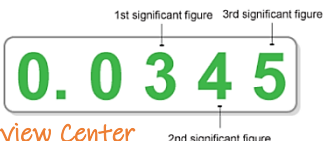




Significant Figures

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Significant figures or significant digits are the meaningful digits in a measurement. It is usually impossible to get the exact value of a quantity being measured, hence it is important to indicate the margin of error by indicating which digits are significant. Increasing the number of significant figures makes for a more accurate measurement though this usually depends on the type of measuring device being used.

Rules of Significant Figures

- Non-zero digits are significant.
- Zero between non-zero digits are significant.
- Zeros to the left of the first non-zero digit are not significant.
- If a number is greater than 1, all zeroes after the decimal point are considered significant.
- If a number is less than 1, zeroes that are after and between the non-zero digits are considered significant.
- For numbers with no decimal points, zeroes after the last non-zero digit may or may not be significant.

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0.00062100

Zeros are not significant after decimal before non-zero numbers

All nonzero numbers are significant

Zeros after nonzero numbers in a decimal are significant



Accuracy & Precision

The degree of accuracy tells us that the measurements are close to the exact value while the degree of precision tells us of the consistency of the measurements. Accurate measurements are usually precise while precise measurements are not necessarily accurate. Usually, the devices or instruments used in measuring the different parameters affect the accuracy and precision of the measurement.

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Low accuracy, but high precision
systematic error

High accuracy, high precision
good

Low accuracy, low precision
bad

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Error & Percent Error

Error is the difference between the accepted value and the measured value.

Percent error, on the other hand, is the ratio of an error to an accepted value.

$$\text{Error} = \text{Accepted value} - \text{Measure value}$$

$$\% \text{Error} = \frac{|\text{Error}|}{\text{Accepted value}} \times 100$$



The Atomic Theory

Even before the formulation of the Atomic Theory by the English scientist, John Dalton, the Greek philosopher Democritus already proposed the idea that all forms of matter can be divided into tiny indivisible particles in the 5th century BC. Democritus called these particles as "atomos", which means "indivisible".

John Dalton's

Atomic Theory can be summarized using the following points.



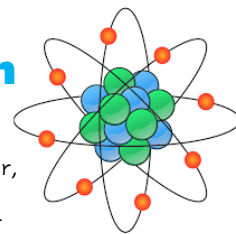
- Atoms make up an element, which is a substance that can no longer be decomposed into a simpler substance by chemical means. The atoms of one element are identical while atoms of different elements are also different.
- Combining different atoms from two or more elements will make up a compound. When atoms combine to form compounds, they follow the **Law of Definite Proportions**. This law states that when two elements combine to form a distinct chemical compound, they combine with a definite ratio of masses that is characteristic of the combining elements and the compound that is formed. This was first proposed by Joseph Louis Proust.

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- A chemical reaction can occur spontaneously without creation and disappearance of new atoms. This is called the **Law of Conservation of Mass**.

The Atom

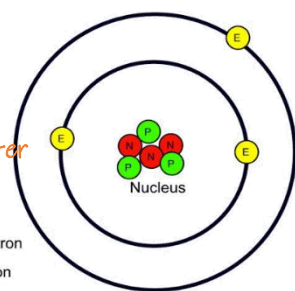
The atom is the basic unit of all elements. However, the atom can be subdivided further into smaller components that we refer to as subatomic particles. The three subatomic particles are the electron, proton and the neutron.



Inside the Atom using the Bohr Model of the Atom

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Electron
Proton
Neutron



The atomic model used today (Bohr Model) describes the electron as subatomic particles that surround the extremely small nucleus, which contains the protons and neutrons. The mass of the whole atom is concentrated in the nucleus since the mass of the proton and mass of the neutron are 1836 and 1840 times the mass of the electron, respectively (mass of electron is negligible relative to the mass of the whole atom). However, in terms of the volume, the nucleus occupies only about 1/1013 of the volume of the whole atom. In other words, the atom consists of a central core which we call as the nucleus wherein the mass is concentrated and a large empty space where electrons can be found.

	Electron	Proton	Neutron
Discoverer	Sir JJ Thomson	Ernest Rutherford	James Chadwick
Country of Birth	England	New Zealand (Although he spent most of his life in England)	England
Charge	-1.6×10^{-19} C	$+1.6 \times 10^{-19}$ C	0
Unit Charge	-1	+1	0
Mass (kg)	9.109×10^{-31}	1.6725×10^{-27}	1.6749×10^{-27}
Mass (u or amu)	0.00548 u 0.00548 amu	1.007276 u 1.007276 amu	1.008665 u 1.008665 amu
Year Discovered	1897	1911	1932
Nobel Prize for the Discoverer	1906	1908 (For his work on radioactivity)	1935

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